

CONCENTRATION, NOT TARGETING: A PRE-REGISTERED NULL FOR UNCERTAINTY-GUIDED SYNTHETIC DATA IN CAMOUFLAGED OBJECT DETECTION

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ABSTRACT

A control that returns zero is informative only if it could have returned something else. We report two cases in camouflaged object detection where the default control certifies the comforting answer. First, allocating a synthetic-data budget by model uncertainty appears to select a markedly different training set than random selection — but selection rules that keep the allocation’s *shape* while destroying the uncertainty signal reproduce that separation entirely, and the signal itself adds +0.0073 of a Cohen’s *d* over its own shuffle. In 24 pre-registered training runs the targeted budget does not beat the random one, and in 12 further runs holding concentration exactly fixed, real, destroyed and reversed targeting are indistinguishable in 12/12 cells. Second, exact hashing reports 0 collisions between CHAMELEON and COD10K-train, while re-encode-tolerant matching finds that 41 of CHAMELEON’s 76 images (53.9%) are the same photographs as training data; we claim non-independence rather than inflation, and release a detector and clean protocol. We also find that across four uncertainty-signal variants and two architectures, uncertainty predicts pixel error better than structural error in 8/8 pre-registered tests, and that this survives controlling for area. Both nulls are bounded by sensitivity, which we quantify rather than assert.

1 INTRODUCTION

Camouflaged object detection is label-scarce: annotating an object whose appearance statistics match its surroundings is slow and expensive. A generative model that can synthesise camouflaged images offers a way out, and once generation is cheap a natural question follows — if the budget is finite, where should it be spent? The answer the field’s methods propose is appealing: partition the unlabelled target distribution into clusters, score each by how much the model disagrees with its own moving average, and allocate generation in proportion, so new data arrives where the model is weakest. Every component is standard, each is cheap, and the composite is exactly the kind of closed-loop system that ought to work.

The thesis: the default control certifies the comforting answer. Testing that idea, we twice found the standard instrument returning a clean zero where a better instrument does not. Comparing a targeted allocation against a *random* one is the obvious control, and it credits to targeting an effect that concentration alone produces; the control that discriminates is a same-shape allocation with the uncertainty signal destroyed. Checking an evaluation benchmark for training contamination by *exact hashing* is the obvious control, and it certifies as perfectly clean a benchmark of which over half is training data; the control that discriminates is re-encode-tolerant matching. In both cases the default returns 0, the result is the one everybody would prefer, and it is wrong. That pattern, not either result alone, is what this paper is about.

What we did, and the scope, in the same breath. Across 24 pre-registered training runs — two architectures, four arms, three seeds — a targeted allocation does not beat a random one: the decisive gap is WITHIN NOISE on both architectures and both endpoints under a rule fixed before

the first run. That campaign’s bar is 126% of the improvement the approach reports, so it resolves coarsely; a second campaign of 12 runs tightens it by $3.24\times$, and under *that* bar the decisive gap sits at $0.91\times$ threshold — still within noise, and marginally so. Both are therefore *no effect resolvable at this sensitivity*, never a demonstration of absence. To show the harness can resolve anything at all, we add a mean-teacher positive control.

Contributions. Each maps to one section; the tiers say how far each travels. **Tier 3** findings are independent of the method, **Tier 2** concern the target data or the uncertainty signal, **Tier 1** are properties of this loop.

- **[Tier 3]** 41 of CHAMELEON’s 76 images (53.9%) are re-encoded COD10K-train and CAMO training images, while exact hashing returns 0. Verified against an author-sourced release and calibrated by a negative control finding 0 of 1082 checkable NC4K images colliding with the COD10K test split. We claim non-independence rather than inflation (Section 7).
- **[Tier 2]** The target distribution has no cluster structure to allocate over — best silhouette 0.1600, no better in three embedding spaces (Section 6.1).
- **[Tier 2]** An uncertainty signal can contribute nothing beyond the concentration it induces: $+0.0073$ of a Cohen’s d against its own shuffle, and with concentration held *exactly* fixed, real, destroyed and reversed targeting are indistinguishable in 12/12 cells (Section 6.2).
- **[Tier 2]** Uncertainty predicts pixel error better than structural error across four signal variants and two architectures, 8/8 under a criterion fixed in advance, surviving a pre-registered area control (Section 6.3).
- **[Tier 1, scoped]** Why this instantiation could not have shown a gain: a pinned optimisation budget, a foreground supply that is a bijection onto 4447 copied objects, and a generated distribution narrower than its own input. We also *withdraw* one of our own candidate causes (Section 6.4). None licenses the conclusion that closed-loop generation cannot work.

Decision rules were committed before the runs they govern, at named commits; amendments are append-only and dated; thresholds that failed are reported failed, and two that *passed* are reported and then refused.

2 RELATED WORK

Camouflaged object detection and its benchmarks. The task asks for pixel-accurate segmentation of objects whose appearance statistics match their surroundings [CITE: COD task definition / survey], reported on a small fixed set of benchmarks — COD10K [CITE: COD10K benchmark], CAMO [CITE: CAMO benchmark], NC4K [CITE: NC4K benchmark] and CHAMELEON [CITE: CHAMELEON benchmark] — with search-and-identification architectures [CITE: SINet], [CITE: SINet-v2], [CITE: SegMaR] and structure-aware measures [CITE: S-measure], [CITE: E-measure], [CITE: weighted F-measure], since pixel-average error rewards the background majority. That those four constitute the standard reporting set is what makes Section 7 consequential beyond any single method.

Synthetic data and generative augmentation. Generating training data with diffusion models is a standard response to label scarcity in dense prediction [CITE: diffusion-generated training data for segmentation], [CITE: latent diffusion], [CITE: DDIM], with compositing-based augmentation its cheaper predecessor [CITE: copy-paste augmentation]. For camouflage, a synthetic-to-real line [CITE: synthetic-to-real domain adaptation] regenerates a background around a real foreground [CITE: LAKE-RED] and couples that generator to a self-training loop [CITE: S2R-COD]. That pairing is what we test. Other camouflage generators exist [CITE: other camouflage image generation] and we do not test them, so our null is about this instantiation.

Uncertainty-guided acquisition. The idea under test is reasonable, and prior work is why. Acquiring data where a model is least certain is long-established [CITE: uncertainty sampling / active learning survey], with disagreement between a model and its moving average [CITE: mean teacher] or among ensemble members [CITE: deep ensembles] and predictive entropy [CITE: predictive

entropy / MC-dropout] as standard estimators, and consistency training [CITE: consistency regularisation / FixMatch] and pseudo-labelling [CITE: pseudo-labelling] as the usual way to exploit unlabelled data. In the batch setting acquisition is also spread over a representation, by clustering or core-set selection [CITE: core-set / diversity-based acquisition], here over self-supervised features [CITE: DINOv2], [CITE: CLIP]. Uncertainty and diversity are known to be entangled in batch acquisition [CITE: batch active learning, uncertainty-diversity trade-off]; Section 6.2 measures how much benefit survives when the uncertainty is destroyed and only the diversity remains.

Benchmark integrity and negative results. Train-test contamination has repeatedly been found in established vision benchmarks [CITE: train-test contamination in vision benchmarks], [CITE: CIFAR/ImageNet duplicate analysis], and is a first-order concern in evaluating large models [CITE: benchmark contamination in LLM evaluation]; our contribution is that exact hashing certifies a contaminated set as clean. We follow arguments for pre-registration [CITE: pre-registration in machine learning] and for publishing disconfirming results [CITE: negative results / reproducibility in ML], and our finding that a learned probe separates two encodings of identical content joins existing criticism of learned distributional distances [CITE: critiques of learned-probe distributional distance], [CITE: FID / representation-metric critiques].

3 THE APPROACH UNDER TEST

A negative result is only as informative as the specification of what was tested. We therefore give the loop in full, and state which of its properties are pre-registered, which are inherited from the released implementation, and which we measured rather than assumed.

3.1 SETTING AND LOOP

The task is source-to-real domain adaptation for camouflaged object detection. The labelled source pool holds 4447 image and ground-truth pairs; the unlabelled target pool holds 4040 images and no labels. The primary endpoint is the COD10K test split, the secondary NC4K. The target pool is not one distribution: it mixes 3040 images from COD10K with 1000 from CAMO, and a held-out probe separates those components at up to 0.9225 AUC, so a method partitioning this pool is partitioning a mixture — a finding, not furniture, and Section 6.1 gives the consequence.

Training runs in two rounds. Round one trains a student against the source pool while an exponential-moving-average teacher supplies a consistency target on the unlabelled images. Between rounds, target images whose student–teacher edge disagreement falls below a fixed fraction of the set average are appended with the teacher’s prediction as their label. Round two rebuilds both models from ImageNet initialisation and trains on the enlarged pool; its best teacher checkpoint is evaluated. Two properties matter and neither is visible from “closed loop”: the round count is a fixed constant, not a convergence criterion, and the generator is never invoked from training — it renders its pool once, offline, and the only edge returning from training to data carries pseudo-labels, not images. It composites rather than synthesises, regenerating a background around a given foreground and copying the original object pixels back (Section 6.4).

3.2 UNCERTAINTY-GUIDED ALLOCATION

The idea under test spends a fixed generation budget where the model is currently weakest, in four stages.

(i) Clustering. The target pool is partitioned by k -means into $k = 75$ clusters over L2-normalised self-supervised embeddings, fitted once and never refitted. **(ii) Deficiency scoring.** Each image receives a scalar from the student–teacher disagreement — the quantity the consistency loss already optimises, computed without ground truth, which is what makes it available on unlabelled data:

$$\text{ES}(x) = a \|\nabla \sigma(s(x)) - \nabla \sigma(t(x))\|_1 + b \text{BCE}(\sigma(s(x)), \sigma(t(x))), \quad (1)$$

with student and teacher logits s, t , sigmoid σ , Sobel gradient magnitude ∇ , and configured weights $a = 0.9, b = 0.3$; the per-cluster score es_j is the mean over its members. **(iii) Softmax allocation.** Budget B is spread by a temperature softmax over es_j with $T = \alpha \cdot \text{sd}(es)$, making α scale-free across spaces and k ; integer budgets follow by largest remainder, summing to B exactly. We use

$\alpha = 1.0$, $B = 1000$. (iv) **Greedy foreground selection.** Each funded cluster draws images by cosine similarity between its centroid and a grey-masked cutout of each candidate foreground — the cutout, because at selection time the foreground exists and the render does not. Appendix A gives the remaining specification.

3.3 CONDITIONS

Four arms share one base pool and differ only in what is appended. **A0**, the unpadded baseline, is the base pool of 4447 pairs alone — the row comparable to published numbers, and an *unclean* control, differing in pool size as well as content. **A2** adds the 1000 original photographs of arm B’s stems, separating pool size from generated content. **B** adds 1000 renders drawn uniformly at random, redrawn per seed. **C10** adds the 1000 renders chosen by the allocation above. A2, B and C10 all hold 5447 pairs, and the pre-registered claim is the gap between C10 and B.

What is held constant is verified rather than assumed: across all 24 runs the base pool is asserted byte-identical against a hash manifest with 0 mismatches, both rounds read the unfiltered 4040-image target set, and every run shares one trainer invocation, checkpoint rule, endpoint set and deterministic-kernel setting. Section 4 adds two arms holding this allocation’s shape fixed while varying only the signal’s direction.

4 EXPERIMENTAL SETUP AND PRE-REGISTRATION

Every decision rule in this paper was committed before the runs it governs, and we state them here, before any result, because a null is only worth reading if the reader can tell it was not produced by choosing how to look.

Endpoints and seeds. The primary endpoint is COD10K-test S_α from the final round’s best teacher checkpoint. NC4K S_α is secondary and, in the pre-registration’s words, *reported, never decides*; MAE, F_β^w and E_ϕ decide nothing. Seeds are {42, 43, 45}. CHAMELEON is not an endpoint, and the reason is itself a result of this paper (Section 7).

The decision rule. Let $\hat{\sigma}$ be the pooled within-arm standard deviation of S_α across arms, at 8 degrees of freedom. The rule, frozen before the first run:

REAL EFFECT iff $\Delta > 2\hat{\sigma}$ and sign consistent 3/3 seeds
 WITHIN NOISE iff $|\Delta| \leq 2\hat{\sigma}$
 REAL REGRESSION iff $\Delta < -2\hat{\sigma}$ and sign consistent 3/3 seeds
 INCONCLUSIVE iff $|\Delta| > 2\hat{\sigma}$ but sign not 3/3 — report as is, do *not* add seeds

The rule is applied independently per architecture — SINet primary, SINet-v2 the robustness check, neither overriding the other, a disagreement reported as a disagreement. At $n = 3$ we report no p -value, bootstrap or multiple-comparison correction: a design choice, not an omission. Verdicts are Δ against a $2\hat{\sigma}$ bar with sign-consistency counts, and nothing converts them into significance claims.

Two further arms that isolate direction from shape. Arms C10 and B differ by *concentration* as well as by targeting, so that comparison alone cannot speak to the uncertainty signal. A second, separately pre-registered campaign of 12 runs — two architectures $\times$ three arms $\times$ three seeds — holds the allocation’s shape fixed and varies only the signal’s direction. **CSHUF** permutes the per-cluster scores across clusters, destroying targeting; **CINV** reverses their rank, making the allocation maximally anti-targeted. CSHUF’s permutation is the first of 64 pre-declared draws satisfying zero fixed points and $|\rho| \leq 0.10$, chosen by a rule fixed before the draw rather than by inspecting its consequences; its realised rank correlation with the true score vector is -0.03351 , against -1.0 for CINV. Because equality between arms is the *supporting* outcome here, two near-identical arms would manufacture it, so set distinctness was gated before any GPU time at a Jaccard bound of 0.50: between 452 and 511 of each pair’s 1000 injected images differ.

Provenance of the rules. Each rule is recorded at a named commit preceding the runs it governs, and amendments are append-only and dated. The sharpest is a frozen clause requiring reproduction

Table 1: The pre-registered verdict on COD10K-test S_α , three seeds per arm. $\hat{\sigma}$ is pooled within-arm at $df = 8$; the frozen rule declares WITHIN NOISE when $|\Delta| \leq 2\hat{\sigma}$. The architectures disagree on the sign of the decisive gap. Arm means, per-arm standard deviations and all per-run values at full precision are in Table 6. *Source:* ABC.RESULTS.md §1.3–§1.4; EXP ABC #3 @ results/REBUILD_LOG.txt:1376.

	$\hat{\sigma}$ ($df = 8$)	$2\hat{\sigma}$	$\Delta(B - A2)$	$\Delta(C - B)$	Verdict
SINet	0.0090	0.0179	+0.0058 3/3	+ 0.0050 3/3	WITHIN NOISE
SINet-v2	0.0061	0.0123	+0.0030 3/3	- 0.0004 2/3	WITHIN NOISE

of a committed metric table to better than 10^{-9} , which no correct computation could satisfy because the file records six decimal places. Appendix A gives the commits and all amendments. It is not presented as minor: the author of that rule also authored its flaw.

5 THE PRE-REGISTERED NULL

Under the rule of Section 4, applied to 24 training runs, $\Delta(C - B)$ is WITHIN NOISE on both architectures and both endpoints (Table 1); the secondary endpoint returns the same verdict on every gap.

The power statement, and the sensitivity analysis it demands. This campaign resolved more coarsely than it declared. Its bar, $2\hat{\sigma} = 0.0179$, is *larger* than this repository’s own reproduction gap of 0.0142 (mean teacher 0.7030 to the full method 0.7172) that it was pre-registered as able to half-resolve — 126% of the effect it was built to detect. The corresponding published gap is 0.0152; we use the reproduction figure throughout because it is the one measured on this stack.

The falsification campaign estimates a bar $3.24\times$ tighter, $\hat{\sigma} = 0.0028$ against 0.0090, and because its noise pool contains arms B and C10 — the two arms of the decisive gap — that bar legitimately applies here. **Applying it:** $\Delta(C10 - B) = 0.0050$ **against a bar of 0.0055 is $0.91\times$ threshold, with 3/3 sign consistency. Still WITHIN NOISE, and marginally so.** We state the asymmetry rather than leave it to be found: where equality is the *supporting* outcome a tighter bar is conservative, but for this null the claim and the null coincide, so the looser bar is anti-conservative. The conservative estimator was available where it cost us nothing, and we have now applied it where it costs us something. Even so it resolves only 39% of the reference gap on SINet and 45% on SINet-v2. Appendix G places these arms against the published numbers, including A0’s reproduction gate.

The aggregate pattern, which points away from our conclusion. The ordering $A0 < A2 < B \approx C10$ is monotone in all four cells, three of the four steps sign-consistent 3/3. If real it would mean padding the pool helps and targeting adds nothing on top. The frozen rule does not convert it into a verdict — no step clears its bar, and at $n = 3$ we pre-registered no test that aggregates across cells — but our own standard forbids suppressing a pattern for failing a threshold, and this one does not flatter us.

A threshold that passed, and was refused. On the secondary endpoint, $\Delta(A0 \rightarrow C10)$ for SINet is +0.0109 with sign consistent 3/3, against a bar of 0.0083: the frozen rule returns REAL EFFECT. We do not report it as one, on two independent grounds each sufficient by itself. NC4K is the secondary endpoint and decides nothing, and the comparison is against A0, which is both confounded in pool size and the least stable arm in the campaign. We record it prominently *because* it is the tempting one.

6 WHAT GENERALIZES: A TIERED DISSECTION

The null does not establish that the approach fails, only that no benefit is resolvable at this sensitivity; it says nothing about the uncertainty signal, since C10 and B differ by concentration as well as targeting; and its scope is inherited from the generator. Those three gaps are what this section

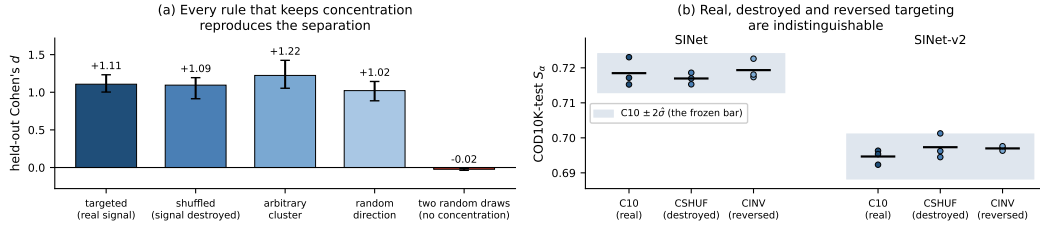


Figure 1: **(a)** Held-out Cohen’s d between a selected set and a random set of equal size. Every rule preserving the allocation’s *concentration* reproduces the separation — including permuting the uncertainty scores across clusters, and ignoring them entirely; only destroying concentration collapses it. Bars are means over four cells, whiskers the range. **(b)** Trained S_α for the three arms holding concentration *exactly* fixed while varying only the signal’s direction; points are seeds, rules are arm means, the band is the frozen $\pm 2\hat{\sigma}$ bar. *Source:* `cl_attribution.csv` (EXP C1 #4); `out/t2/abc_metrics.csv` (EXP T2 #3).

measures. We separate causes by scope: **Tier 2** findings concern the target data or the uncertainty signal and travel beyond this instantiation, **Tier 1** findings are properties of this loop and do not. Table 7 in Appendix H assigns every cause.

6.1 THE TARGET DISTRIBUTION HAS NO CLUSTER STRUCTURE TO ALLOCATE OVER

[Tier 2] Cluster-wise allocation presumes clusters; this pool barely has them. Its best silhouette is 0.1600 at $k = 75$ in the space the pipeline uses, and the other two spaces reach only 0.1465 and 0.0568 — all below any conventional threshold, so the weakness is not an artifact of one encoder. The third fails more specifically: its silhouette falls monotonically from 0.0568 at $k = 5$ to 0.0357 at $k = 150$, so the criterion returns its smallest grid value by hitting the boundary rather than finding a maximum. Our pre-declared guard required an interior maximum; it failed, and we record the failure rather than relax it. Two measurements reinforce this: an arbitrary preprocessing choice reassigns 5.4% of cluster memberships, so an effect smaller than that cannot be attributed to targeting; and the pool is the two-dataset mixture of Section 3.1, separating from itself at up to 0.9225. **Scope:** three embedding spaces and k -means only. **Transfers:** this check is upstream of any signal question and costs one silhouette sweep.

6.2 AN UNCERTAINTY SIGNAL CAN CONTRIBUTE NOTHING BEYOND THE CONCENTRATION IT INDUCES

[Tier 2] Our most transferable result, and the first instance of this paper’s thesis; Figure 1 carries it.

A targeted selection separates from a random one of equal size at a held-out Cohen’s d of roughly 1.00 to 1.23, an order of magnitude above a previously claimed 0.10. That separation is real and is not caused by targeting: control arms keeping the allocation’s *shape* while destroying the signal reproduce it — permuting scores across clusters gives +0.9139 to +1.1403 in the same cells, and an *arbitrary* cluster as much again; only destroying concentration too collapses it. Paired across twenty cells the signal adds +0.0073 of a d against its own shuffle, winning 13/20 — weak evidence for a small positive contribution ($P \approx 0.13$ under a fair coin), not the coin flip we previously called it — and is *negative* at -0.0649 against an arbitrary cluster, winning 4/20 ($P \approx 0.006$). Applying the same yardstick both ways, the strongest directional evidence here points *against* targeting. What the targeted arm does is narrow: an effective-rank ratio of 0.634 on average over 20/20 cells, 0.53 to 0.64 in the four peak cells, covering no more of the target manifold.

Geometry is not performance. The falsification arms hold concentration *exactly* fixed — CSHUF permutes the per-cluster scores across clusters and CINV reverses their rank, both reproducing the committed allocation’s shape to five decimal places on all four recorded shape statistics — and vary only direction. All twelve cells are WITHIN NOISE; the largest gap anywhere is 0.0050 at $1.82\hat{\sigma}$. A directional pattern exists and we report it without claiming it: C10 is the best arm in none of the four

cells, the anti-targeted CINV has the highest mean in three of four, and the architectures disagree in sign on two of three gaps — the signature of noise. None of it reaches the bar, so it is not the verdict.

Scope, and the trained half is the weaker one. Negligible *in this geometry* is supported; zero and harmful are not. The trained comparison is attenuated three times over and the factors multiply: injected images are 1000 of a 5447-image pool (18.4%); the optimisation budget is pinned, so added data dilutes exposure; and only $\approx 48\%$ of injected images differ between arms — an effective contrast near 8.8% of training content against a bar of 0.0055. Stated plainly: **the 12/12 trained result is weakly informative, and the geometric result carries this claim.** The attenuation biases toward the null, which is exactly why it must be disclosed. **Transfers:** the control that matters for uncertainty-guided acquisition is not random selection but a same-shape allocation with the signal destroyed.

6.3 UNCERTAINTY PREDICTS PIXEL ERROR BETTER THAN STRUCTURAL ERROR

[Tier 2] Under a criterion fixed before measurement, the ordering $\rho(\text{MAE}) > \rho(1 - S_\alpha) > \rho(1 - \text{IoU})$ holds on 8 of 8 whole-image rows — four signal variants across three families (student-teacher disagreement, predictive entropy, and ensemble disagreement in two variants) crossed with two architectures — seven of them at all ten clustering seeds. The gap spans +0.0708 to +0.3330; all clear the pre-registered resolution floor of about 0.06, but the smallest only just does.

The same criterion demonstrably can fail, which is what makes its passing informative: aggregated over a boundary band, only 2 of 8 rows pass and four go negative, outside our pre-registered interpretation space, which did not anticipate inversion. We report it as it fell.

The area control, pre-registered after that failure and before it was run. Our own explanation for the boundary collapse — a whole-image mean is uncertainty density times area — *applies to the whole-image rows too*: if MAE and the signals both scale with area while S_α carries an object-size term that does not, the ordering could be a property of the metrics. We therefore partialled area out on the same partition and floors, with the consequence fixed in advance, using two covariates: object area on the endpoint side and uncertain-region area on the target side. The ordering survives 7/8 and 6/8 against a floor of 6/8 declared before the run — **area-robust, with the second covariate clearing by exactly zero margin.** Three details cut against the simple reading: partialling object area *raises* $\rho(1 - S_\alpha)$ in all eight rows (+0.3433 to +0.4988 for ES on SINet), the opposite of the confound’s prediction; the single object-area failure is a tie, crossing by 0.0012; and both uncertain-area failures are on SINet-v2, where $\rho(\text{MAE})$ falls +0.7070 to +0.4457 once band area is removed. **Area accounts for part of the gap but not its sign.**

Scope: correlational, one endpoint, and the *magnitude* does not generalise — $\rho(1 - S_\alpha)$ reaches +0.5628 on SINet-v2, so the claim is strictly ordinal and the boundary question remains unreachable. **Transfers:** if you care about structural quality, an uncertainty signal of these kinds is a better proxy for pixel-average error than for what you want to improve.

6.4 WHY THIS INSTANTIATION COULD NOT HAVE SHOWN A GAIN

[Tier 1] Three properties of *this* loop and *this* generator explain why the campaign could not have demonstrated a gain, and none licenses the conclusion that closed-loop generation cannot work. The optimisation budget is pinned at 253 and 127 steps per epoch in both rounds of every run regardless of pool size, so adding 1000 images bought no additional optimisation — only a change of mixture. The foreground supply is a bijection onto the base set: all 4447 renders trace into it, 0 outside, and of the 8885 objects with a valid interior comparison (4443 + 4442 across the two pools, not 2×4447), 0 show any sign of regeneration — the objects are copied, not synthesised. And the generated distribution is narrower than its own input: real photographs of an entirely different genre cover the target manifold better than the synthetic images generated from those very photographs, a paired loss of -23.6% to -82.9% . That last measurement is the only evidence here not bounded by statistical power.

Each carries a legitimate dismissal, which we state rather than await: the first is a property of the training schedule, not a result about synthetic data; the second follows from choosing a generator that composites by design; the third is a distributional characterisation, so “the distribution is far,

Table 2: Which evaluation columns can be reported as independent measurements. Every rate is a *lower bound*: only same-dimension images are comparable, so an image with no candidate is unchecked, not clean. Crops, rescales and different photographs of one specimen are neither detected nor claimed. In the exact-hash column 0 is a measured zero, “—” not applicable. *Source*: CLEAN_PROTOCOL.md; EXP D2 #5, EXP D2R #4, EXP D2_NC4K #2.

Endpoint	Contaminated	Exact hash	Unchecked	Nearest-neighbour gap	Verdict
COD10K-test	2/2026 (0.1%)	—	524/2026	none	reportable [†]
NC4K	1/4121 (0.0%)	—	2406/4121	none	reportable [†]
CHAMELEON	41/76 (53.9%)	0	25/76	41 below 5.51, next 40.58	not reportable
CAMO	4/250 (1.6%)	—	155/250	none	selection set only

[†] Measured against on-disk copies, not author-sourced ones — the same weakness the CHAMELEON re-audit closes. That re-audit is *owed* for these rows; until then their rates are provisional.

therefore no amount of it can help” would be a non-sequitur. We also *withdraw* one of our own candidate causes: the convenient explanation was a narrow conditioning channel, and reading the generator’s architecture refutes that as stated, since four routes carry foreground information and the dominant one bypasses the summary. **This is resolved by source reading, not by measurement**, and showing the channel is not narrow does not show its capacity is used — so we decline both the claim that the generator is not the binding constraint and that it is steerable. Appendix C gives the full measurements.

7 BENCHMARK CONTAMINATION AND A CLEAN PROTOCOL

This is the paper’s thesis again, in its cleanest form: the default control returns 0, the answer is the one everybody would prefer, and it is wrong. Exact hashing is how dataset overlap is normally checked; it reports the benchmark below as perfectly clean. It is not. Nothing here depends on whether the method of Section 3 worked.

The finding. 41 of CHAMELEON’s 76 images — 53.9% — are the same photographs as images in the COD10K-train and CAMO training pool, re-encoded; 40 lie in the public COD10K-train split and one in CAMO. A model trained on COD10K-train has therefore already seen over half of CHAMELEON before being evaluated on it. We verified against an author-sourced release rather than our own copy, byte-identical at 76/76, so the result does not depend on which copy was audited. CHAMELEON predates COD10K, so the parsimonious reading is that COD10K-train absorbed these images when assembled — but only pool membership is measured, and the direction is inference from chronology.

Exact hashing reports this set as perfectly clean. It returns 0 collisions between CHAMELEON and the training pool, and is correct at that level: these are re-encodings, not byte-identical copies. The number was right; the inference drawn from it was not. We hit the same trap twice — on prior work, and on our own first implementation, which reported 10/76 (13.2%), roughly a quarter of the true count, because contrast normalisation destroyed the discriminative scale. The first wrong number was ours.

The count is a property of the data, not of a tolerance. Matches at mean absolute difference $\leq 1, 2, 3, 5, 6$ are 11, 26, 37, 40 and 41 of 76: the count saturates. Independently, the sorted nearest-neighbour distances place 41 images below 5.51 with the next at 40.58 — a $7.36\times$ jump after the forty-first — unchanged whether we verify the top 8, the top 32, or every candidate. Two instruments that owe each other nothing land on the same integer, and the other three endpoints show no discontinuity.

A negative control, from the same instrument. An audit that finds contamination wherever it looks is worthless. The identical detector — imported, not re-implemented — returns a clean null on NC4K against the COD10K test split: 0 byte-identical, 0 pixel-identical, 0/4121 re-encoded at

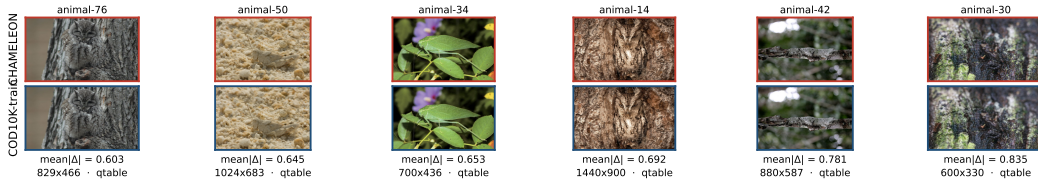


Figure 2: Six of the 41 confirmed pairs: the author-sourced CHAMELEON image (top) and its partner in the COD10K-train pool (bottom). *Selection rule, applied not curated*: the six lowest mean absolute difference among pairs with a COD10K-train partner. Exact hashing reports 0 collisions for every pair shown, because each is a re-encoding rather than a byte-identical copy. *Source*: `chameleon_contaminated.json` (EXP D2R #4).

every tolerance swept, no gap, and a minimum nearest distance of 20.610, $3.44\times$ the confirmation tolerance. NC4K’s 1/4121 in Table 2 is measured against the *full training pool* — a different axis from the 0/4121 against COD10K-train and against COD10K-test, and the three must not be collapsed.

What we measured for, and could not report. We expected to report by how much a contaminated column misleads. We measured for it and found nothing: across both mask releases all eight difficulty percentiles for the leaked subset land within 0.448 to 0.559, and the sign of the clean-minus-leaked difference *flips* with the release. The pre-declared reading — a percentile inside $[0.25, 0.75]$ means no skew — is what stopped us writing the quotable sentence. Our claim is therefore non-independence, not inflation: *CHAMELEON is not an independent endpoint for COD10K-trained models*.

The protocol: retire the benchmark, do not subset it. Our released protocol originally recommended reporting CHAMELEON on its 35 uncontaminated images. We withdraw that here, because our own numbers do not support it: of 76 images, 41 are contaminated and 25 more have no same-dimension candidate and are therefore *unchecked, not clean*, leaving **only 10 verifiably clean**. A 10-image endpoint is not an endpoint, and with two incompatible mask releases the honest recommendation is to retire CHAMELEON rather than subset it — stated as a change to a released artifact rather than a silent tightening. Report COD10K-test and NC4K instead, disclosing the unchecked share. We release the detector alongside this paper. It is checkable without our data: it passes an eight-assertion synthetic known-answer self-test using no repository data, reproduces the canonical 41 pairs and 41 names exactly, and surfaced one additional pair — the second partner of the one image that has two. [TODO: verify — the authors state that the published S2R-COD paper reports a CHAMELEON row in its Table 3, which would place that specific column within scope of this finding; the PDF is not in the checkout, so this is not verifiable from a committed artifact.] [TODO: author finalize — anonymised release URL for the detector, the contaminated-pair list, and the clean protocol.]

8 CONCLUSION

Instantiated in this loop with this generator, uncertainty-guided closed-loop generation yields no accuracy gain resolvable at 39% to 45% of the improvement the approach reports — a null bounded by sensitivity and scoped to an exhausted foreground supply. What survives concerns the data and the signal rather than our loop: the target distribution does not decompose into clusters any allocation method could spend a budget over, the uncertainty signal contributes nothing beyond the concentration it induces, and every uncertainty signal we could compute predicts pixel-average error better than the structural error this task is scored on. A working system would need an optimisation budget that grows with the data, genuinely new foregrounds, a boundary-aware signal and a steerable generator. Both halves of this paper are then one lesson: a control that returns 0 is only as good as its ability to return something else, and twice the field’s default cannot. Both return the comforting answer; the remedy in each case was a control built to fail.

AI USE STATEMENT

[TODO: author finalize — this statement is required by ICLR 2027 and must describe the authors’ actual use of generative AI tools. It must be written by the authors, not inferred, and it should state which tasks with required disclosure such tools were and were not used for, which are not applicable, and how AI-assisted work was verified. One verification mechanism is available to this work and should be named if relevant: every numeric value in this paper is traceable to a committed log block emitted by a committed script, and no result was transcribed by hand; the traceability map in Appendix B records the mapping. The authors take responsibility for the final content of this work.]

ETHICS STATEMENT

This work involves no human subjects and no personally identifying data. It uses public research benchmarks for camouflaged object detection under their published terms [TODO: author confirm — dataset licence statements for each benchmark used].

Section 7 reports an overlap between two public datasets. We state explicitly what that measurement does and does not support. It is a measurement of pool membership; it attributes no intent to any dataset’s authors, and we draw the direction of provenance as an inference from release chronology rather than as a finding. We measured for a score-inflation figure, did not find one, and do not report one. The one correction we direct at anyone is directed at ourselves: it was our own protocol that adopted CHAMELEON as an evaluation endpoint without checking whether the repository we built on ever sanctioned it. We release the detector so that the measurement can be repeated, contested, or extended by others rather than taken on trust.

REPRODUCIBILITY STATEMENT

The decision rules governing every result are committed before the runs they govern, at the commits named in Section 4, and amendments to them are append-only and dated. Each experiment has a committed result document and one or more committed log blocks; each number in this paper is traceable to such a block or to a committed artifact, and Appendix B gives that mapping. Per-run metrics for all 36 training runs, the full clustering sweep, and the full signal-generalizability table are in the appendix. The contamination detector is released with an eight-assertion known-answer self-test that runs without any of our data [TODO: author finalize — anonymised URL for code, artifacts, and the released detector].

Two limits on reproducibility are worth stating rather than discovering. Our regeneration check establishes byte-identical reproduction on one machine with one software stack; it is not a claim of determinism across hardware or driver versions. And a render from this generator is a function of its foreground, its mask, *and its position within the processing shard*, so changing the shard count changes the outputs even at a fixed seed.

A METHOD SPECIFICATION

Training. SINet at batch 16 for 40 epochs and SINet-v2 at batch 32 for 100 epochs, Adam at 10^{-4} , images at 352×352 with ImageNet normalisation. The EMA teacher uses momentum 0.996 under this task setting. The consistency weights in Equation 1 are the configured $a = 0.9$, $b = 0.3$ with the unweighted cross-entropy variant — note that these are the values the trainer sets for this task, not the loss class’s own defaults. Strong views add random autocontrast and Gaussian blur to the weak view. `cudnn.deterministic` is enabled and `cudnn.benchmark` disabled; this reduces, and is not claimed to eliminate, kernel nondeterminism.

Optimisation budget. Steps per epoch are set by the shorter of the source and target loaders, which fixes them at 253 (SINet) and 127 (SINet-v2) in both rounds of every run regardless of pool size. This is measured per run and asserted, and it is the mechanism behind Section 6.4’s first cause.

Allocation sweeps. The budget grid is $B \in \{250, 500, 1000, 2000, 3000, 4447\}$ and the temperature grid over α is extended until both degenerate ends are reached, defined numerically as concen-

trated when $\max_j p_j \geq 0.95$ and uniform when the total variation from uniform falls below 0.01, with caps at $\alpha \in [10^{-4}, 10^3]$. The campaign uses $\alpha = 1.0$, $B = 1000$.

Metrics. S_α at $\alpha = 0.5$ is the primary endpoint metric; MAE is the checkpoint-selection metric; F_β^w and E_ϕ are computed and reported but decide nothing. The F -measures apply β in the non-squared form, so the metric name rather than a formula is what identifies them.

The A2 control’s one confound. A2 pairs the authors’ original photographs with the authors’ masks, whose mean foreground fraction differs from the raw masks by 0.0058 on 1000 of 5447 images. This affects $\Delta(B - A2)$ and is disclosed; it does not touch $\Delta(C10 - B)$, the decisive gap, since B and C10 share mask provenance, pool size and render pool.

Pre-registration commits and amendments. The four-arm rule is recorded at 065dac6, the commit carrying that campaign’s first log block; the falsification campaign’s at 617a2e1, before its first training run; the signal-generality check’s at 6efde9f; the area control’s at c2114af; the positive control’s at c2114af. Three amendments exist, all append-only and dated. Two are reference-tolerance defects of the same kind: a frozen clause requiring reproduction to better than 10^{-9} against a file recording six decimal places, and — in the area control, written by the same author who had just documented the first — the identical mistake against a file recording four. Both are replaced by exact agreement at the recorded precision, which is the strictest satisfiable form, and neither altered a decision threshold. The third marks one verification gate not applicable in scope after it was found to compare two architectures’ signals.

B SUPPORTING TABLES

Table 3: Full clustering sweep over the target set: silhouette (mean over 10 k -means seeds), bootstrap ARI and seed ARI, for every k in the declared grid and all three embedding spaces. The principled k per space is **bold**. No k in any space reaches a silhouette conventionally regarded as well-separated structure, and clipL224 falls monotonically, so its selection criterion returns the grid edge rather than an interior maximum — recorded as a FAIL of the declared guard. *Source:* rebuild/B1/out/bl_k_sweep-{dinoL224,dinoL518,clipL224}.csv; block EXP B1 #3 @ results/REBUILD-LOG.txt:922.

k	dinoL224			dinoL518			clipL224		
	sil.	bs.ARI	sd.ARI	sil.	bs.ARI	sd.ARI	sil.	bs.ARI	sd.ARI
5	0.0538	0.712	0.795	0.0566	0.624	0.732	0.0568	0.953	0.983
10	0.0868	0.676	0.709	0.0923	0.598	0.770	0.0550	0.692	0.805
15	0.1053	0.608	0.714	0.1152	0.573	0.592	0.0486	0.575	0.670
20	0.1180	0.629	0.693	0.1326	0.506	0.524	0.0486	0.635	0.646
30	0.1378	0.596	0.642	0.1527	0.539	0.617	0.0488	0.609	0.631
50	0.1465	0.610	0.615	0.1556	0.579	0.600	0.0448	0.498	0.549
75	0.1457	0.588	0.627	0.1600	0.579	0.616	0.0396	0.463	0.477
100	0.1419	0.608	0.637	0.1557	0.583	0.601	0.0382	0.419	0.452
150	0.1367	0.630	0.636	0.1507	0.618	0.625	0.0357	0.380	0.409

C TIER 1 CAUSES IN FULL

[Tier 1] These causes are properties of this loop and this generator. They explain why this attempt could not have demonstrated a gain; they do not license the conclusion that closed-loop generation cannot work. Each carries a legitimate dismissal, which we state.

The optimisation budget is pinned at 253 and 127 steps per epoch for the two architectures, in both rounds of every run, regardless of pool size — so adding 1000 images to a 4447-image pool bought no additional optimisation, only a change of mixture. A reviewer may call this a property of the training schedule rather than a result about synthetic data, and that reading is correct: nothing here

Table 4: Full signal-generality measurement, all 16 rows: four signal variants across three signal families, two architectures, two aggregations. ρ at the committed seed, $\pm$ sd over 10 k -means seeds. The whole-image block passes the pre-registered ordering 8/8; the boundary block passes 2/8 with four rows going negative, an outcome outside the pre-registered interpretation space, reported as it fell. *Source*: rebuild/T2C/out/t2c_table.csv; block EXP T2C @ results/REBUILD_LOG.txt:2550.

arch	signal	agg.	$\rho(\text{MAE})$	$\rho(1 - S_\alpha)$	$\rho(1 - \text{IoU})$	order	seeds
SINet-v2	ES	whole	$+0.7070 \pm 0.0404$	$+0.5628 \pm 0.0359$	$+0.4591 \pm 0.0315$	PASS	10/10
SINet-v2	ensemble A0	whole	$+0.6191 \pm 0.0675$	$+0.5483 \pm 0.0409$	$+0.4540 \pm 0.0451$	PASS	9/10
SINet-v2	ensemble CSHUF	whole	$+0.5576 \pm 0.0690$	$+0.4488 \pm 0.0580$	$+0.3370 \pm 0.0577$	PASS	10/10
SINet-v2	entropy	whole	$+0.7393 \pm 0.0309$	$+0.5404 \pm 0.0408$	$+0.4070 \pm 0.0395$	PASS	10/10
SINet-v2	ES	bound	-0.5485 ± 0.0745	-0.1146 ± 0.0675	$+0.0537 \pm 0.0558$	FAIL	0/10
SINet-v2	ensemble A0	bound	-0.1960 ± 0.0524	$+0.0425 \pm 0.0410$	$+0.0933 \pm 0.0396$	FAIL	0/10
SINet-v2	ensemble CSHUF	bound	-0.2527 ± 0.0668	-0.0591 ± 0.0609	-0.0011 ± 0.0579	FAIL	0/10
SINet-v2	entropy	bound	$+0.4208 \pm 0.0356$	$+0.4182 \pm 0.0455$	$+0.3605 \pm 0.0470$	PASS	8/10
SINet	ES	whole	$+0.6284 \pm 0.0381$	$+0.3433 \pm 0.0301$	$+0.2613 \pm 0.0383$	PASS	10/10
SINet	ensemble A0	whole	$+0.6028 \pm 0.0466$	$+0.2698 \pm 0.0487$	$+0.1491 \pm 0.0560$	PASS	10/10
SINet	ensemble CSHUF	whole	$+0.5212 \pm 0.0564$	$+0.4006 \pm 0.0459$	$+0.3361 \pm 0.0447$	PASS	10/10
SINet	entropy	whole	$+0.5892 \pm 0.0368$	$+0.4267 \pm 0.0277$	$+0.3588 \pm 0.0351$	PASS	10/10
SINet	ES	bound	-0.0012 ± 0.0481	-0.0567 ± 0.0500	-0.0554 ± 0.0479	FAIL	8/10
SINet	ensemble A0	bound	$+0.3633 \pm 0.0605$	$+0.2910 \pm 0.0509$	$+0.2215 \pm 0.0513$	PASS	10/10
SINet	ensemble CSHUF	bound	$+0.3175 \pm 0.0637$	$+0.3857 \pm 0.0381$	$+0.3770 \pm 0.0322$	FAIL	2/10
SINet	entropy	bound	$+0.2434 \pm 0.0414$	$+0.2492 \pm 0.0540$	$+0.2432 \pm 0.0563$	FAIL	6/10

Table 5: The area control (EXP AC). First-order partial Spearman on the same cluster partition and floors as the committed measurement. z_{obj} = per-cluster mean object area on the endpoint side; z_{unc} = per-cluster mean uncertain-region area on the target side. Ordering survives 7/8 and 6/8 against a floor of 6/8 declared before the run. Note that partialling object area *raises* $\rho(1 - S_\alpha)$ in every row — the opposite of the confound’s prediction. *Source*: rebuild/AC/out/ac_table.csv; block EXP AC @ results/REBUILD_LOG.txt:2612.

arch	signal	raw		$ z_{\text{obj}} $		$ z_{\text{unc}} $	
		MAE	$1 - S_\alpha$	MAE	$1 - S_\alpha$	MAE	$1 - S_\alpha$
SINet	ES	+0.6284	+0.3433	+0.5761	+0.4988	+0.5518	+0.3897
SINet	entropy	+0.5892	+0.4267	+0.5505	+0.5517	+0.5060	+0.4426
SINet	ensemble A0	+0.6028	+0.2698	+0.5411	+0.4301	+0.5603	+0.2583
SINet	ensemble CSHUF	+0.5212	+0.4006	+0.5002	+0.4857	+0.4857	+0.3937
SINet-v2	ES	+0.7070	+0.5628	+0.7062	+0.6272	+0.4457	+0.4744
SINet-v2	entropy	+0.7393	+0.5404	+0.7134	+0.6546	+0.5369	+0.4266
SINet-v2	ensemble A0	+0.6191	+0.5483	+0.6245	+0.5969	+0.4355	+0.4488
SINet-v2	ensemble CSHUF	+0.5576	+0.4488	+0.5344	+0.5209	+0.4482	+0.3615
ordering holds		8/8		7/8		6/8	

measures what happens when the optimisation budget grows with the data, which is the largest open question this paper creates.

The foreground pool is exhausted; the background is not. The render set is a bijection onto the base foreground set — all 4447 renders trace into it, 0 outside, 0 base foregrounds unrendered — on the pool training reads as well as on ours, and the mapping is literal: interior error against the source foreground is 5.603 and 9.633 mean absolute grey levels against background error near 70.6, and of 8885 traced objects 0 show any sign of regeneration. This is an identity statement about pixels, not a claim about similarity, which is what makes it resistant to the obvious counter-argument. The dismissal — this generator composites by design, so of course it adds no new foregrounds — is right, and what survives it is the scope statement every null here inherits: evidence about targeting

Table 6: Per-run COD10K-test metrics for every training run in the two frozen campaigns, at full recorded precision. Arms B and C10 appear once; they were re-scored, not re-trained, for the falsification campaign and reproduced their committed values exactly at six decimal places. *Source:* rebuild/ABC/out/abc_metrics.csv and out/t2/abc_metrics.csv; blocks EXP ABC #3 and EXP T2 #3.

arch	arm	seed	S_α	MAE
SINet	A0	42	0.715086	0.074960
SINet	A0	43	0.681706	0.077044
SINet	A0	45	0.706058	0.077347
SINet	A2	42	0.705507	0.075994
SINet	A2	43	0.708334	0.078489
SINet	A2	45	0.709196	0.080858
SINet	B	42	0.711554	0.076085
SINet	B	43	0.713632	0.076278
SINet	B	45	0.715155	0.075917
SINet	C10	42	0.717165	0.073184
SINet	C10	43	0.723034	0.075136
SINet	C10	45	0.715243	0.078355
SINet	CINV	42	0.722612	0.073910
SINet	CINV	43	0.717343	0.077323
SINet	CINV	45	0.718084	0.073607
SINet	CSHUF	42	0.718602	0.074840
SINet	CSHUF	43	0.716974	0.077678
SINet	CSHUF	45	0.715291	0.079340
SINetv2	A0	42	0.688813	0.091691
SINetv2	A0	43	0.678740	0.095147
SINetv2	A0	45	0.699883	0.085756
SINetv2	A2	42	0.691914	0.089186
SINetv2	A2	43	0.688977	0.094524
SINetv2	A2	45	0.695414	0.087455
SINetv2	B	42	0.698340	0.092242
SINetv2	B	43	0.689481	0.089883
SINetv2	B	45	0.697385	0.087132
SINetv2	C10	42	0.692321	0.093395
SINetv2	C10	43	0.696328	0.090749
SINetv2	C10	45	0.695360	0.089182
SINetv2	CINV	42	0.697008	0.088438
SINetv2	CINV	43	0.696331	0.087293
SINetv2	CINV	45	0.697605	0.090189
SINetv2	CSHUF	42	0.701269	0.087506
SINetv2	CSHUF	43	0.694490	0.087305
SINetv2	CSHUF	45	0.696218	0.089485

under an exhausted foreground supply, silent on whether a larger or more diverse one would behave differently.

The generated distribution is narrow, and this evidence alone does not depend on power. As recall against the target manifold the synthetic pools reach 0.1277 to 0.5421 across three embedding spaces, while the generator’s own input pool — real photographs of a different genre — reaches 0.7097 to 0.7473 and a different real camouflage dataset reaches 0.7921 to 0.8995: real photographs of the wrong genre cover the target better than synthetic images generated from those very photographs, a paired loss of -23.6% to -82.9% . Effective rank falls 243.3 to 193.2 while precision rises 0.6490 to 0.6886 — narrowing, not displacement. This is a distributional characterisation, not a training-utility bound; reading it as “the distribution is far, therefore no amount of it can help” would be a non-sequitur, and what generalises is the design, not the number.

One candidate cause we removed rather than kept. A convenient explanation was available: that foreground information reaches the regenerated background only through a narrow low-dimensional summary. Examining the released generator’s architecture refutes that account as stated — four routes carry foreground information into the regenerated region, the narrow summary is one, and

the dominant route bypasses it entirely. *This is resolved by source reading, not by measurement*: no experiment was run and no metric logged, and it withdraws one of our own candidate causes rather than establishing a result. Showing the channel is not narrow does not show its capacity is used, so we decline both the claim that the generator is not the binding constraint and that it is steerable.

D CONTAMINATION: CORRECTIONS TO OUR OWN WORK

Corrections to our own work. An author-sourced re-audit reproduced 8 of 8 of our earlier measurements; its three mismatches correct our prior work, not the data. One declared threshold is left *failing* rather than reworded: we required every confirmed pair to differ in its JPEG quantization table and compared with an inequality that scored a *missing* table as a differing one — absence of evidence read as evidence — because two files are PNGs carrying a `.jpg` extension. Corrected, 41 of 41 pairs still carry independent re-encoding evidence, 40 by quantization table and one by container format. Separately, CHAMELEON was never a sanctioned endpoint in the repository we built on; it was our own plan that promoted it, so the framing here is not that we found a defect in someone else’s evaluation.

E THE ENDPOINT-MEASURED SIGNAL

A correction to our own measurement. Our first measurement correlated this signal *on the labelled endpoint* against endpoint error — not the quantity the pipeline allocates by. Measured on the signal actually available, correlation with pixel error falls by $+0.2100$ and $+0.2470$, from $+0.8695$ and $+0.8754$ to $+0.6595$ and $+0.6284$. A signal validated where labels exist overstates what is usable where they do not.

F THE TWO CHAMELEON MASK RELEASES

A second, unrelated reason the column is not comparable. The two circulating CHAMELEON mask releases are different annotations: opposite polarity, mean IoU 0.6932 after alignment, only 27 of 76 identical, and on identical predictions the choice of release alone moves MAE by $2.7\times$. This has nothing to do with contamination, or with our method.

G WHERE THESE ARMS SIT AGAINST THE PUBLISHED NUMBERS

Where these arms sit relative to the published numbers. The published configuration corresponding to C10 reports $S_\alpha = 0.7136$; our C10 measures 0.7185 . A0 is the *unpadded* baseline, not that configuration: its pre-declared reproduction gate was on seed 42, measuring 0.7151 against this repository’s committed 0.7172 , inside the declared $3\sigma = 0.0107$. The A0 arm *mean* of 0.7010 sits lower because A0 is the least stable arm ($\sigma = 0.0173$), which we report as unexplained.

H THE TIERED CAUSE SUMMARY

Table 7: Every cause, its measured quantity, and its scope tier. Tier 2 causes concern the target data or the uncertainty signal and travel beyond this instantiation; Tier 1 causes are properties of this loop or this generator. The tier column, not the magnitude column, determines how far a row may be carried. *Sources, by row:* FINAL_RESULTS.md Tables 8.2, 8.1, 9.2, 9.4, 11.1, 12.1, 5.1, 7.1 and 10.4; blocks EXP B1 #3 and #4, EXP C1 #4, EXP T2 #3, EXP T2C, EXP ABC #2, EXP D1, EXP A3, in results/REBUILD_LOG.txt. The conditioning row has no log block by construction.

Cause	Measured quantity	Tier	Generalizes to
No cluster structure to allocate over	best silhouette 0.1600; 0.1465, 0.0568 elsewhere	2	any method allocating over clusters of such a distribution
Membership is not a robust label	5.4% of memberships move under a preprocessing change	2	a floor beneath which an effect cannot be attributed to targeting
Signal adds nothing beyond concentration, in geometry	+0.0073 of a d against its own shuffle, 13/20	2	any uncertainty-guided acquisition: the shuffle is the control
... and nothing resolvable on trained accuracy	12/12 WITHIN NOISE, 3.24 $\times$ tighter bar	2	this signal class carries no resolvable directional information
Targeting buys proximity, not coverage	effective-rank ratio 0.53–0.64, 20/20	2	concentration narrows the set rather than extending its reach
Pixel error predicted better than structural error	ordering passes 8/8 rows, 3 signals $\times$ 2 architectures	2	uncertainty-guided allocation for this task in general
Endpoint-measured signal overstates itself	ρ drops +0.2100, +0.2470	2	any signal validated where labels exist
Optimisation budget is pinned	253 / 127 steps, both rounds, every run	1	nothing — a property of this loop
Foreground pool is exhausted	bijection 4447/4447; 0 outside, 0 unrendered	1 $\rightarrow$ 2	the scoping sentence any comparable null must carry
Objects are copied, not generated	interior 5.603 / 9.633 vs background 70.802 / 70.555	1	the measurement design, reusable on any inpainting generator
Generated distribution is narrow	recall 0.13–0.54 vs 0.71–0.90 for every real set	1	the design: a paired within-content coverage delta
Conditioning channel is <i>not</i> the bottleneck	four routes; source reading, no experiment	1	nothing — a withdrawn cause